

rachel dudley, p.e.

principal, sireneum

architectural designer licensed engineer (electrical), AZ (#67301)

design@sireneum.com | 970-380-9333

design portfolio: sireneum.com

practice focus

strengthening relationships between humans and their places.

technical skills

Computational Design:

Revit, Rhino, Grasshopper, QGIS, ClimateStudio, Adobe Creative Suite, Lumion

Human AI-Collaborative Design:

Iterative concept development, research synthesis, design critique, visualization, and technical communication feedback and assistance.

experience

Sireneum Designs (2021-Present)

independent design practice integrating architecture, engineering, visualization, and research through residential and community-scale projects

Principal

TEAM Integrated Engineering/BB&E (2010-2021)

GEITA Contractor supporting the Air Force Civil Engineering Center

Electrical Engineer — Misawa, Japan/Remote

education

Master of Architecture (M.Arch, 2026)

four time design excellence award winner, distinguished graduate

Arizona State University — Tempe, AZ

Electrical Engineering (B.S.E., 2009) *power + signal processing*

capstone design team award winner, summa cum laude

Arizona State University — Tempe, AZ

Business Administration (B.S., 2003)

honors program, summa cum laude, minor in communication

Regis University — Denver, CO

community engagement

Designing Sustainable Solutions with Communities: Rim Country Learning Landscape (2026)

Community Science for Global Futures Certificate, Arizona State University

City of Litchfield Park (2018-Present)

Board of Adjustment (Chair) · Design Review Board (Vice Chair)

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selected writing

particle:wave:: Architecture at the Threshold of Resonance — Critical essay examining the role of architecture through the lens of particle-wave duality, arguing that the human experience is fundamentally both discrete and continuous and should be treated as such in architectural practice (included as a sample of applied systems analysis).

selected work

Outside-In House (2026)

AOR: Devan Porter, Principal, Design Service Professionals

residential remodel in central Phoenix organized around an Eichler-inspired central courtyard

Lead Designer / Project Manager

The Divide (2025)

Instructor: Arne Emerson, Partner, Morphosis

capstone studio exploring disaster, environmental systems, and adaptive public architecture

Studio - Unbuilt

Here (2024)

Instructor: Karín Santiago, Principal, Lightvox Studio

studio design with mixed-use residential and museum program exploring uncertainty and scarcity the desert

Studio - Unbuilt

ISIAC (2024)

Instructor: Ian Dickenson, Associate Principal, Sasaki

multi-generational educational facility exploring community wealth through creativity and making

Studio - Unbuilt

selected references (contact info available upon request)

Ian Dickenson, AIA

Associate Principal, Sasaki

Devan Porter, AIA

Principal, Design Service Professionals

Claudio Vekstein, AIA

Program Head & Professor of Architecture, Arizona State University

Karín Santiago, AIA, LEED AP BD+C

Principal, Lightvox Studio